# GHAZI UNIVERSITY

DERA GHAZI KHAN

*Knowledge • Character • Progress*

*Learn Today Lead Tomorrow*

## ASSIGNMENT Lab




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|---|---|
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| Roll No: | 06 |
| Section: | (C) |
| Submitted To: | Mr Muhammad Asad |
| Assignment: | Lab |







*Better Skills → Brighter Future*

## Lab:

### Task 1

Write a program to input the following variables and display them:

```
int age;
double pay;
char section;
```

## Program:

```
#include <iostream>
using namespace std;

int main()
{
    int age;
    double pay;
    char section;

    cout << "Enter age: ";
    cin >> age;

    cout << "Enter pay: ";
    cin >> pay;

    cout << "Enter section: ";
    cin >> section;

    cout << "\nAge: " << age << endl;
    cout << "Pay: " << pay << endl;
    cout << "Section: " << section << endl;

    return 0;
}
```

## Output:

```
Enter age: 18
Enter pay: 25000.50
Enter section: A

Age: 18
Pay: 25000.50
Section: A
```

## Lab:

# Task 2

Complete the following table by writing the value of each expression in the Value column according to C++ language rules.

## Program:

```
#include <iostream>
using namespace std;

int main()
{
    cout << "28 / 4 - 2 = " << 28 / 4 - 2 << endl;
    cout << "6 + 12 * 2 - 8 = " << 6 + 12 * 2 - 8 << endl;
    cout << "4 + 8 * 2 = " << 4 + 8 * 2 << endl;
    cout << "6 + 17 % 3 - 2 = " << 6 + 17 % 3 - 2 << endl;
    cout << "2 + 22 * (9 - 7) = " << 2 + 22 * (9 - 7) << endl;
    cout << "(8 + 7) * 2 = " << (8 + 7) * 2 << endl;
    cout << "(16 + 7) % 2 - 1 = " << (16 + 7) % 2 - 1 << endl;
    cout << "12 / (10 - 6) = " << 12 / (10 - 6) << endl;
    cout << "(19 - 3) * (2 + 2) / 4 = "
         << (19 - 3) * (2 + 2) / 4 << endl;

    return 0;
}
```

## Output:

```
28 / 4 - 2 = 5
6 + 12 * 2 - 8 = 22
4 + 8 * 2 = 20
6 + 17 % 3 - 2 = 6
2 + 22 * (9 - 7) = 46
(8 + 7) * 2 = 30
(16 + 7) % 2 - 1 = 0
12 / (10 - 6) = 3
(19 - 3) * (2 + 2) / 4 = 16
```

Lab:

# Task 3

Assume a program has the following variable definitions and performs the given operation.

## Program:

```
#include <iostream>
using namespace std;

int main ()
{
    int units;
    float mass;
    double weight;

    units = 5;
    mass = 2.5;

    weight = mass x units;

    cout << "Units = " << units << endl;
    cout << "Mass = " << mass << endl;
    cout << "Weight = " << weight << endl;

    return 0;
}
```

## Output:

```
Units = 5
Mass = 2.5
Weight = 12.5
```

## Lab:

# Task 4

Assume that the following variable definitions are given. Write a program to calculate the value of a and display it.

## Program:

```
#include <iostream>
using namespace std;

int main()
{
    int a, b = 2;
    float c = 4.2;

    a = b * c;

    cout << "Value of a = " << a << endl;

    return 0;
}
```

## Output:

```
Value of a = 8
```

## Lab:

# Task 5

Assume that qty and salesReps are both integers. Use a type cast expression to rewrite the following statement so it will no longer perform integer division.

```
unitsEach = qty / salesReps;
```

## Program:

```
#include <iostream>
using namespace std;

int main()
{
    int qty = 10;
    int salesReps = 4;
    double unitsEach;

    unitsEach = static_cast<double>(qty) / salesReps;

    cout << "Units Each = " << unitsEach << endl;

    return 0;
}
```

## Output:

```
Units Each = 2.5
```

## Lab:

### Task 6

Write a program that can be used as a math tutor for a young student. The program should display two random 3-digit numbers, ask the student to add them, and then display the correct answer.

## Program:

```
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

int main()
{
    int num1, num2, answer;
    char key;

    srand(time(0));

    num1 = rand() % 900 + 100;
    num2 = rand() % 900 + 100;

    answer = num1 + num2;

    cout << "Math Tutor\n\n";
    cout << "   " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "-----" << endl;

    cout << "\nPress any key and Enter to check the answer: ";
    cin >> key;

    cout << "\nCorrect Answer:\n\n";
    cout << "   " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "-----" << endl;
    cout << "   " << answer << endl;

    return 0;
}
```

## Output:

```
Math Tutor

    734
  + 268
  -----
Press any key and Enter to check the answer: a

Correct Answer:

    734
  + 268
  -----
   1002
```

6:38 ▶ 51

# Coding C++

Auto saved at 18:38:16

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int age;
    double pay;
    char section;

    cout << "Enter age: ";
    cin >> age;

    cout << "Enter pay: ";
    cin >> pay;

    cout << "Enter section: ";
    cin >> section;

    cout << "\nAge: " << age << endl;
    cout << "Pay: " << pay << endl;
    cout << "Section: " << section << endl;

    return 0;
}
```


6:38
Compile Result
Enter age:

6:39

# Coding C++

RUN MENU

Auto saved at 18:39:33

```cpp
#include <iostream>
using namespace std;

int main()
{
    cout << "28 / 4 - 2 = " << 28 / 4 - 2 << endl;
    cout << "6 + 12 * 2 - 8 = " << 6 + 12 * 2 - 8 << endl;
    cout << "4 + 8 * 2 = " << 4 + 8 * 2 << endl;
    cout << "6 + 17 % 3 - 2 = " << 6 + 17 % 3 - 2 << endl;
    cout << "2 + 22 * (9 - 7) = " << 2 + 22 * (9 - 7) << endl;
    cout << "(8 + 7) * 2 = " << (8 + 7) * 2 << endl;
    cout << "(16 + 7) % 2 - 1 = " << (16 + 7) % 2 - 1 << endl;
    cout << "12 / (10 - 6) = " << 12 / (10 - 6) << endl;
    cout << "(19 - 3) * (2 + 2) / 4 = "
         << (19 - 3) * (2 + 2) / 4 << endl;

    return 0;
}
```

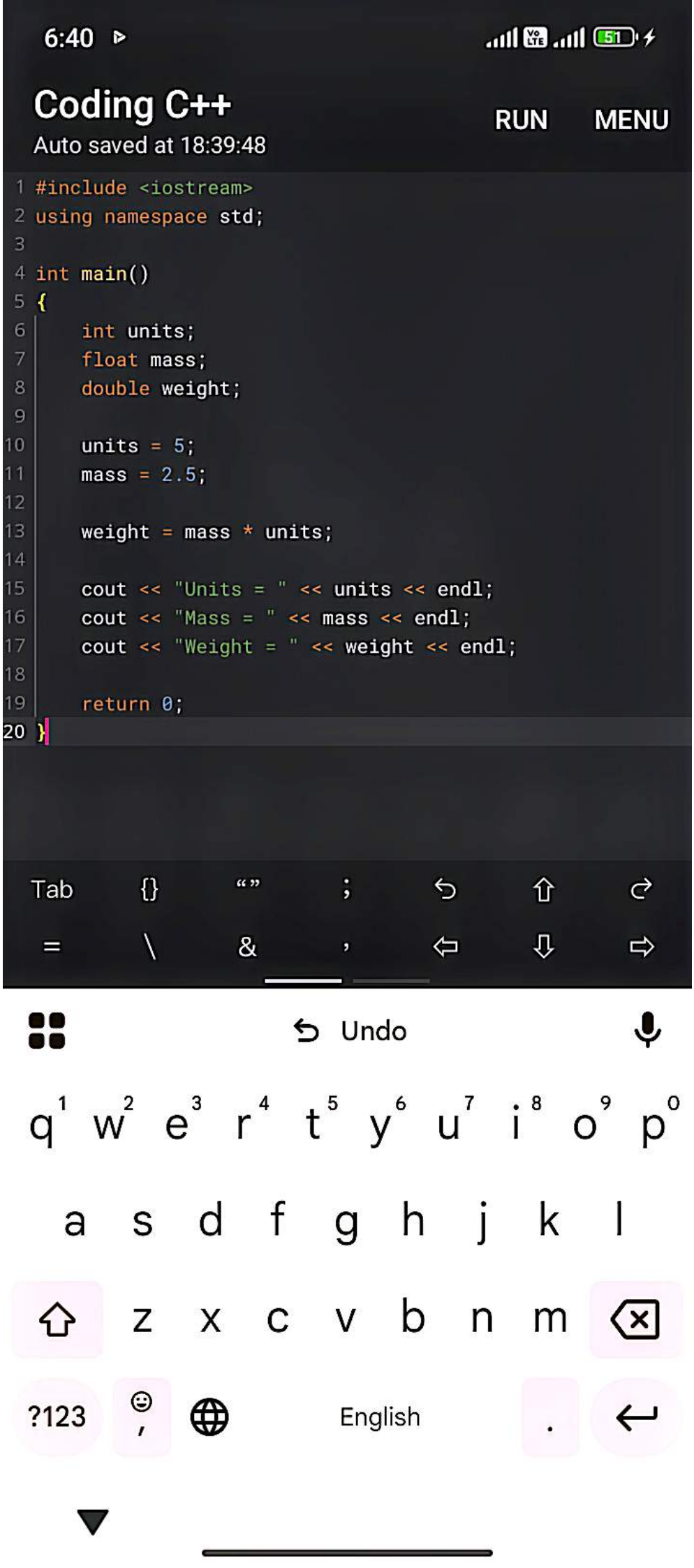
Tab {} “” ; = \ & ,

6:39

# Compile Result

```
28 / 4 - 2 = 5
6 + 12 * 2 - 8 = 22
4 + 8 * 2 = 20
6 + 17 % 3 - 2 = 6
2 + 22 * (9 - 7) = 46
(8 + 7) * 2 = 30
(16 + 7) % 2 - 1 = 0
12 / (10 - 6) = 3
(19 - 3) * (2 + 2) / 4 = 16

[Process completed - press Enter]
```

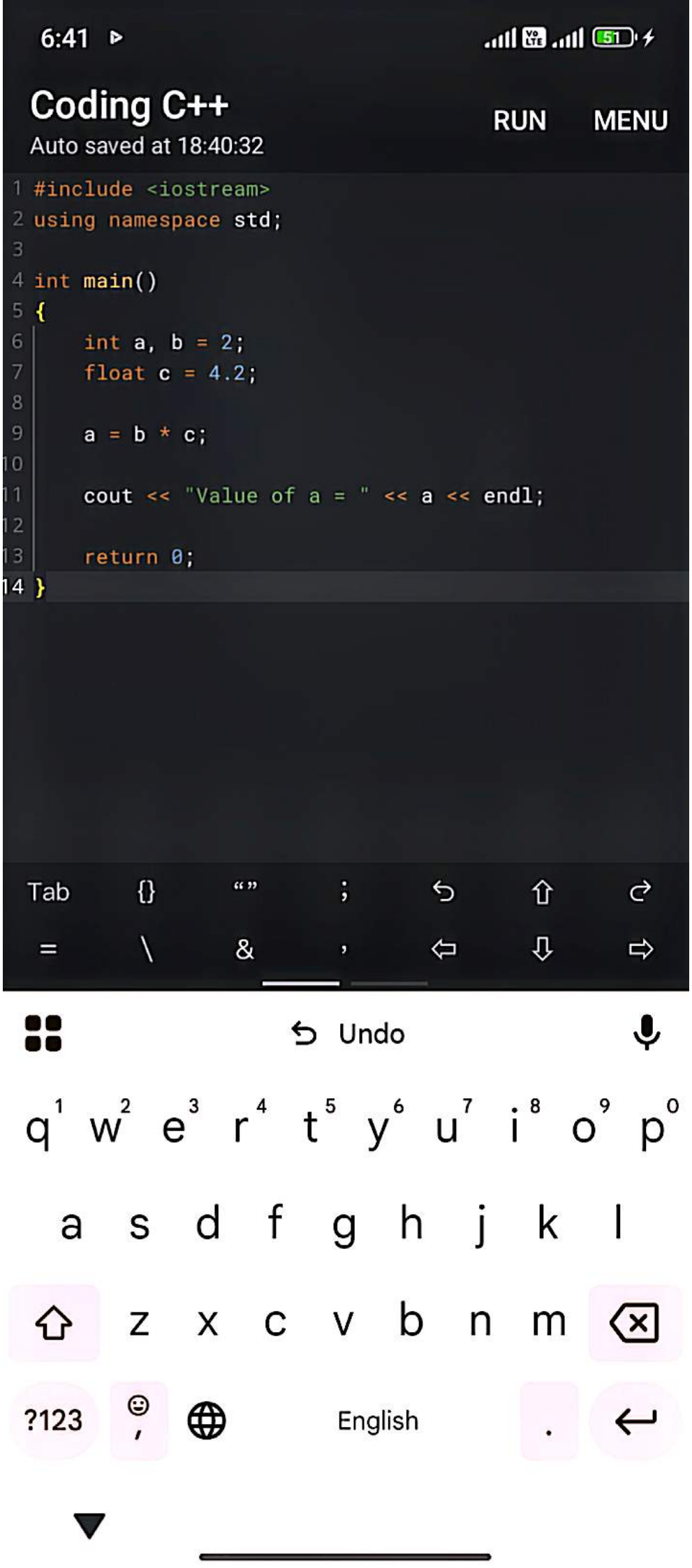

6:40
Coding C++
Auto saved at 18:39:48
RUN
MENU
#include <iostream>
using namespace std;
int main()
{
int units;
float mass;
double weight;
units = 5;
mass = 2.5;
weight = mass * units;
cout << "Units = " << units << endl;
cout << "Mass = " << mass << endl;
cout << "Weight = " << weight << endl;
return 0;
}
Tab
{}
“”
;
=
\
&
,
Undo
English

6:40

# Compile Result

```
Units = 5
Mass = 2.5
Weight = 12.5

[Process completed - press Enter]
```

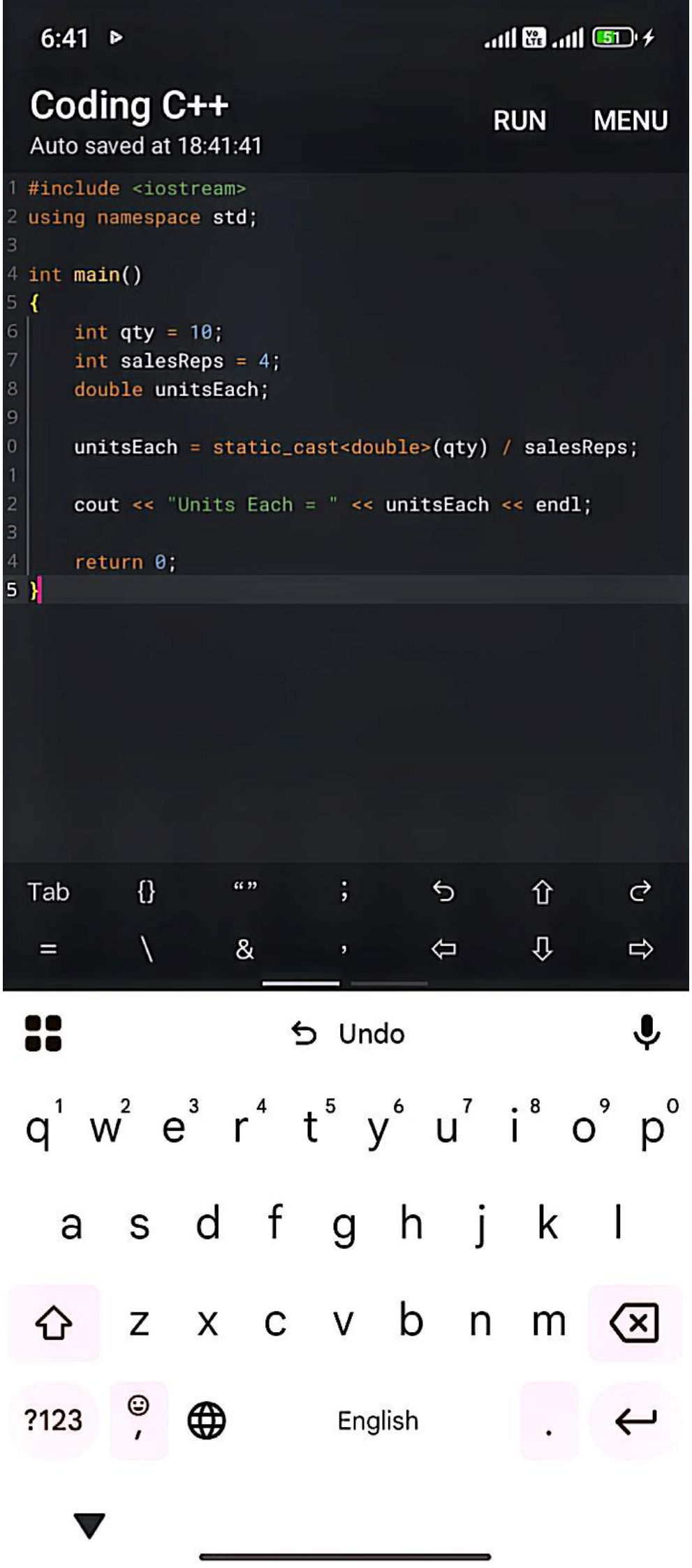

6:41
Coding C++
Auto saved at 18:40:32
RUN
MENU
#include <iostream>
using namespace std;
int main()
{
int a, b = 2;
float c = 4.2;
a = b * c;
cout << "Value of a = " << a << endl;
return 0;
}
Tab
{}
“”
;
=
\
&
,
Undo
English
?123

6:41

## Compile Result

```
Value of a = 8

[Process completed - press Enter]
```


6:41
Coding C++
Auto saved at 18:41:41
RUN
MENU
#include <iostream>
using namespace std;
int main()
{
int qty = 10;
int salesReps = 4;
double unitsEach;
unitsEach = static_cast<double>(qty) / salesReps;
cout << "Units Each = " << unitsEach << endl;
return 0;
}
Tab
{}
“”
;
=
\
&
,
Undo
English
?123

6:41
Compile Result
Units Each = 2.5
[Process completed - press Enter]

```cpp
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;

int main()
{
    int num1, num2, answer;
    char key;

    srand(time(0));

    num1 = rand() % 900 + 100;
    num2 = rand() % 900 + 100;

    answer = num1 + num2;

    cout << "Math Tutor\n\n";

    cout << "   " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "-----\n";

    cout << "\nPress any key and Enter to check the answer: ";
    cin >> key;

    cout << "\nCorrect Answer:\n\n";
    cout << "   " << num1 << endl;
    cout << "+ " << num2 << endl;
    cout << "-----\n";
    cout << "   " << answer << endl;

    return 0;
}
```

# Compile Result

```
Math Tutor

  556
+ 609
-----

Press any key and Enter to check the ans
wer:
```